Coursework 3: Quantification of Hepatic Iron Overload in Thalassemia Patients using MRI

Giuliano Costa **SN: 24225233** **April 15th, 2025**

Abstract

This work addresses the challenge of quantifying hepatic iron overload in thalassemia patients using magnetic resonance imaging (MRI). We develop and validate a robust framework for T2* estimation using total variation regularization within an alternating direction method of multipliers (ADMM) optimization approach. The proposed method is compared against standard Levenberg-Marquardt (LM) fitting algorithms using both synthetic datasets and in-vivo MRI scans from four patients with sickle cell disease. Our results demonstrate that incorporating spatial regularization improves the accuracy and noise-resistance of T2* mapping, particularly in regions with heterogeneous iron distribution or near vascular structures. This non-invasive quantification technique provides a reliable alternative to liver biopsy for monitoring iron accumulation and guiding chelation therapy in transfusion-dependent patients. The methodology presented offers potential clinical value in assessing iron-related complications and treatment efficacy in thalassemia management.

1 Introduction

The liver is a metabolic organ responsible for a number of functions, our work primarily focuses on its detoxification of the blood. Iron overload can be caused in a number of ways, notably Thalassemia, a group of inherited blood disorders that causes substandard erythropoiesis and thus anemia. Thalassemia is thought to have evolved as a protection against Malaria, similar to Sickle Cell disease. Patients with this group of disorders frequently require blood transfusions and this leads to a build up of total body iron, due to the body's inability to excrete iron through waste pathways. Liver iron concentration (LIC) is linearly related to total body iron concentration. Since 1994, non-invasive iron estimation via MRI has become increasingly popular among radiologists, where it was historically assessed using biopsy. Liver biopsies are invasive surgeries that can cause complications for patients and are not always accurate, not to mention being an expensive procedure. Alternatively, iron concentration can be indirectly estimated through measurement of proton signal decay. Reeder et al. reviewed the many existing models and found three methods that are widely used clinically: SIR (signal intensity ratio), T2 relaxometry, and T2* relaxometry. Notably, Resonance Health provides a method called FerriScan for T2 Relaxometry and has been studied extensively. However, T2* mapping has emerged as a preferred technique due to its shorter acquisition times and sensitivity to iron concentration. The fundamental principle of T2* mapping involves measuring signal decay across multiple echo times and fitting this decay to an exponential curve. In patients with iron overload, the presence of iron causes magnetic field inhomogeneities that accelerate the decay of the MR signal, resulting in shorter T2* values. This inverse relationship between tissue iron concentration and T2* values forms the basis for non-invasive iron quantification.

Several mathematical models have been proposed for T2* estimation from MRI data. The simplest approach is the single-exponential model, which assumes a uniform iron distribution within tissue voxels. More complex models include the shifted single-exponential model (adding a constant to account for long T2* components) and the bi-exponential model (incorporating both iron-dense and iron-sparse components). The choice of fitting algorithm also significantly impacts accuracy, with methods ranging from simple linear regression to more sophisticated approaches like the Levenberg-Marquardt (LM) algorithm.

In this work, we develop a robust framework for T2* estimation in hepatic iron overload based on total variation regularization and alternating direction method of multipliers (ADMM) optimization. Our approach addresses limitations of current methods, particularly in handling noise and heterogeneous iron distribution patterns that commonly occur in clinical scenarios. We validate our method against standard techniques using both synthetic data and real patient scans to demonstrate its practical utility in thalassemia management.

2 Related Work

Several approaches for T2* estimation in liver MRI have been proposed in the literature, with the single-exponential model being widely used due to its simplicity and clinical practicality. The Levenberg-Marquardt (LM) algorithm is the most commonly employed fitting method for T2* estimation in clinical practice. Alternative models have been explored to address the limitations of the single-exponential approach, particularly in tissues with heterogeneous iron distribution. These include the bi-exponential model that incorporates both iron-dense and iron-sparse components (Wood and Ghugre, 2008), and the exponential-plus-constant model that accounts for long T2* components. While these more complex models may better represent the underlying tissue characteristics, they often require additional parameters and can be more challenging to fit reliably. Most existing approaches treat each voxel independently during the fitting process, which can lead to increased sensitivity to noise. Our work differs by introducing spatial regularization through total variation constraints while maintaining the widely accepted single-exponential model as the foundation.

3 Methods

3.1 Problem Formulation

The problem of T2* estimation in liver MRI can be formulated as a signal decay fitting problem. For a sequence of T2*-weighted images acquired at different echo times (TE), the signal intensity s at each voxel follows an exponential decay model. In this work, we focus on the single-exponential model which can be expressed as:

$$s = a \exp(-rt) + w \tag{1}$$

where s is the measured signal intensity, a is the signal intensity at TE $= 0$, $r = 1/\mathrm{T2}^*$ is the decay rate, t is the echo time (TE), and w is a noise component typically assumed to be Gaussian. The goal is to estimate the parameters a and r from the observed signal intensities at various echo times.

3.2 Optimization Framework

Traditional approaches to this problem often use least-squares fitting methods such as the Levenberg-Marquardt algorithm. However, these methods treat each voxel independently and do not account for spatial correlations in the liver tissue. We propose an alternative approach based on an optimization framework that incorporates spatial regularization:

$$\text{minimize}_{a,r} \sum_{n=1}^{N} \frac{1}{2} \|s_n - a_n \exp(-r_n t)\|_2^2 + \alpha \mathrm{TV}(a) + \beta \mathrm{TV}(r) \tag{2}$$

where n indexes the voxels, N is the total number of voxels, $\|\cdot\|_2$ denotes the ℓ_2-norm, and TV represents the total variation regularization that promotes spatial coherence while preserving edges. The parameters α and β control the degree of regularization applied to a and r, respectively.

For a vector x representing an image, the total variation is defined as:

$$\mathrm{TV}(x) = \sum_{n=1}^{N} |\nabla_h x_n| + |\nabla_v x_n| = \|Dx\|_1 \tag{3}$$

where ∇_h and ∇_v denote the horizontal and vertical first-order differences at pixel n, $\|\cdot\|_1$ is the ℓ_1-norm, and D is a matrix representing the first-order difference operator.

3.3 ADMM Algorithm for Parameter Estimation

The optimization problem in Equation (2) is convex but non-smooth due to the TV terms. To solve it efficiently, we employ the Alternating Direction Method of Multipliers (ADMM). We introduce auxiliary variables y and z to decouple the data fidelity term from the regularization terms:

$$\begin{aligned} &\text{minimize}_{a,r,y,z} \sum_{n=1}^{N} \frac{1}{2}\|s_n - a_n \exp(-r_n t)\|_2^2 + \alpha\|y\|_{\mathrm{TV}} + \beta\|z\|_{\mathrm{TV}} \\ &\text{subject to} \quad y = a, \quad z = r \end{aligned} \tag{4}$$

The augmented Lagrangian for this problem is:

$$\begin{aligned} L(a, r, y, z, h, g) = &\sum_{n=1}^{N} \frac{1}{2}\|s_n - a_n \exp(-r_n t)\|_2^2 + \alpha\|y\|_{\mathrm{TV}} + \beta\|z\|_{\mathrm{TV}} \\ &+ \frac{\mu}{2}\|y - a - h\|_2^2 + \frac{\mu}{2}\|z - r - g\|_2^2 \end{aligned} \tag{5}$$

where h and g are the scaled dual variables, and $\mu > 0$ is a penalty parameter.

The ADMM algorithm alternates between the following steps until convergence:

$$a^{k+1} = \text{argmin}_a L(a, r^k, y^k, z^k, h^k, g^k) \tag{6}$$

$$r^{k+1} = \text{argmin}_r L(a^{k+1}, r, y^k, z^k, h^k, g^k) \tag{7}$$

$$y^{k+1} = \text{argmin}_y L(a^{k+1}, r^{k+1}, y, z^k, h^k, g^k) \tag{8}$$

$$z^{k+1} = \text{argmin}_z L(a^{k+1}, r^{k+1}, y^{k+1}, z, h^k, g^k) \tag{9}$$

$$h^{k+1} = h^k - (y^{k+1} - a^{k+1}) \tag{10}$$

$$g^{k+1} = g^k - (z^{k+1} - r^{k+1}) \tag{11}$$

3.4 Update Equations

The update for a can be computed analytically:

$$a_n^{k+1} = \frac{u_n^{k\top} s_n + \mu \tilde{y}_n^k}{\tilde{u}_n^k + \mu} \tag{12}$$

where $u_n^k = \exp(-r_n^k t)$, $\tilde{u}_n^k = u_n^{k\top} u_n^k$, and $\tilde{y}_n^k = y_n^k + h_n^k$.

For the r update, since there is no closed-form solution, we use gradient descent:

$$r_n^{p+1} = r_n^p - \delta \frac{\partial H(r_n^p)}{\partial r_n^p} \tag{13}$$

where p indexes the gradient descent iterations, δ is the step size, and the gradient is given by:

$$\frac{\partial H(r_n^p)}{\partial r_n^p} = a_n^{k+1}(t \odot \exp(-r_n^p t))^\top (a_n^{k+1} \exp(-r_n^p t) - s_n) + \mu(r_n^p - z_n^k - g_n^k) \tag{14}$$

where $\odot$ denotes the Hadamard (element-wise) product.

The updates for y and z involve solving TV-regularized problems, which can be efficiently computed using Chambolle's algorithm:

$$y^{k+1} = \text{Chambolle}\left(a^{k+1} - h^k, \frac{\alpha}{\mu}\right) \tag{15}$$

$$z^{k+1} = \text{Chambolle}\left(r^{k+1} - g^k, \frac{\beta}{\mu}\right) \tag{16}$$

3.5 Stopping Criterion

The algorithm terminates when the sum of the primal and dual residuals is below a predefined threshold:

$$\|y^k - a^k\|_2 + \|z^k - r^k\|_2 + \mu(\|h^k - h^{k+\rho}\|_2 + \|g^k - g^{k+\rho}\|_2) \leq \epsilon \tag{17}$$

where $\epsilon = \sqrt{N} \times 10^{-5}$ and $\rho = 10$.

3.6 Implementation Details

The algorithm was implemented in MATLAB. For the initialization, we set the initial values of $a^{(0)}$ and $r^{(0)}$ based on a simple exponential fit using the Levenberg-Marquardt algorithm. The regularization parameters α and β were set to 10^{-3} for all experiments, and the penalty parameter μ was adaptively updated within the algorithm to keep the primal and dual residual norms within a factor of 10 of one another.

For validation purposes, we tested our method on both synthetic datasets with known ground truth and real patient MRI data. In the synthetic experiments, we generated data with varying T2* values and initial intensities, and added Gaussian noise to simulate realistic imaging conditions. For the patient data, we used MRI scans acquired on a 3Tesla Siemens Skyra scanner using an 8-echo gradient echo sequence with TEs ranging from 1 to 16.4ms.

4 Experiments

Our experimental validation consists of several tasks designed to evaluate the performance of our proposed T2* estimation framework for hepatic iron overload quantification.

Task 2 - Fitting an Exponential Model

In this task, we derive and implement an unconstrained minimization problem for fitting the single exponential model using total variation regularization. We convert this problem into a constrained form suitable for solution via ADMM.

Task 3 - Implementation of the Exponential Fitting Model

We implement the ADMM-based algorithm for T2* estimation using MATLAB. See Methods section.

Task 4 - Synthetic Dataset Creation

To validate our approach, we generate controlled synthetic datasets with known T2* values and initial intensities. We create a sequence of 12 images at different echo times.

Task 5 - Testing the Developed Model on Synthetic Data

We evaluate our ADMM-based approach on synthetic datasets with known ground truth values. The synthetic dataset consists of a 64×64 image divided into four quadrants, each with different combinations of initial intensity (a) and T2* values.

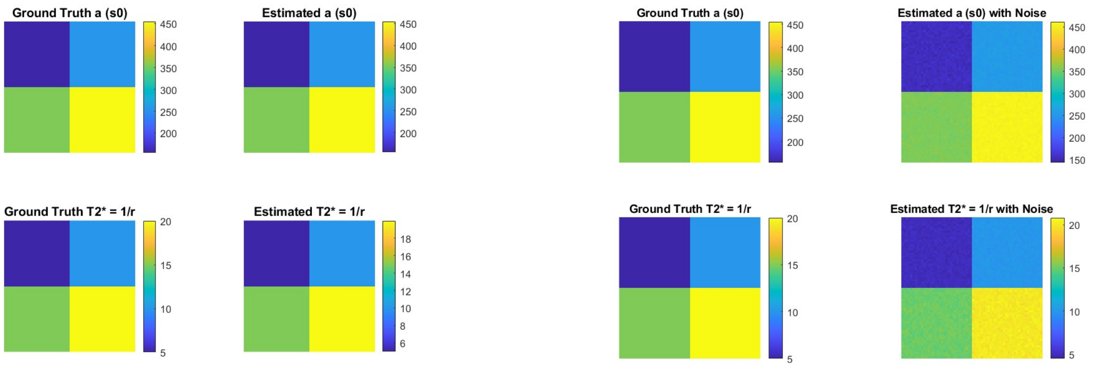
Figure 1 shows the visual comparison of the estimated parameters in both noise-free and noisy conditions.

Our ADMM-based approach demonstrates remarkable accuracy in estimating both the initial intensity (a) and relaxation time (T2*) parameters. In the noise-free scenario, the method achieves near-perfect estimation with errors below 0.11%. Even in the presence of significant noise ($\sigma^2 = 10$), the estimation remains highly accurate with maximum errors of only 0.23%.

The total variation regularization effectively suppresses noise while preserving the sharp boundaries between regions with different T2* values, which is crucial for clinical applications.

Table 1: Comparison of ground truth and estimated parameters for synthetic data

Box	Parameter	Ground Truth	Without Noise	With Noise	Error Range
1	a	155.00	154.83	154.80	-0.11% to -0.13%
	T2* (ms)	5.00	5.01	5.01	0.19% to 0.23%
2	a	255.00	255.00	255.04	0.00% to 0.01%
	T2* (ms)	10.00	10.00	10.00	0.00% to -0.04%
3	a	355.00	355.01	355.06	0.00% to 0.02%
	T2* (ms)	15.00	15.00	15.00	0.00% to -0.01%
4	a	455.00	455.00	454.92	0.00% to -0.02%
	T2* (ms)	20.00	20.00	20.02	0.00% to 0.08%



(a) (b)

Figure 1: Task 5 - ADMM T2* estimation without (a) and with noise (b), noise is a Gaussian with mean zero and variance 10.

Task 6 - Evaluation of Regularization Functions

We investigated the impact of total variation regularization on T2* estimation accuracy by varying the regularization parameters λ_A and λ_R. Table 2 presents the error metrics for selected parameter configurations.

Table 2: Impact of regularization parameters on T2* estimation accuracy

Regularization	λ_A	λ_R	Avg a Error (%)	Avg T2* Error (%)	Max T2* Error (%)
Default	10^{-5}	10^{-5}	0.05	0.09	0.20
Very Weak	10^{-7}	10^{-7}	0.05	0.09	0.20
Moderate	10^{-3}	10^{-3}	0.05	0.09	0.20
Strong	10^{-1}	10^{-1}	0.04	0.08	0.17
Only λ_A Strong	10^{-1}	10^{-5}	0.04	0.09	0.19
Only λ_R Strong	10^{-5}	10^{-1}	0.04	0.08	0.17

The results show that regularization strength has a modest but measurable impact on estimation accuracy. Stronger regularization ($\lambda = 10^{-1}$) slightly improves both average and maximum errors compared to weaker regularization settings. Notably, regularization on the relaxation parameter (λ_R) appears to have a more significant effect on T2* accuracy than regularization on the initial intensity (λ_A) However, the differences across regularization settings remain small in this controlled synthetic dataset, indicating that our approach is not overly sensitive to parameter tuning and can deliver accurate results across a range of regularization strengths.

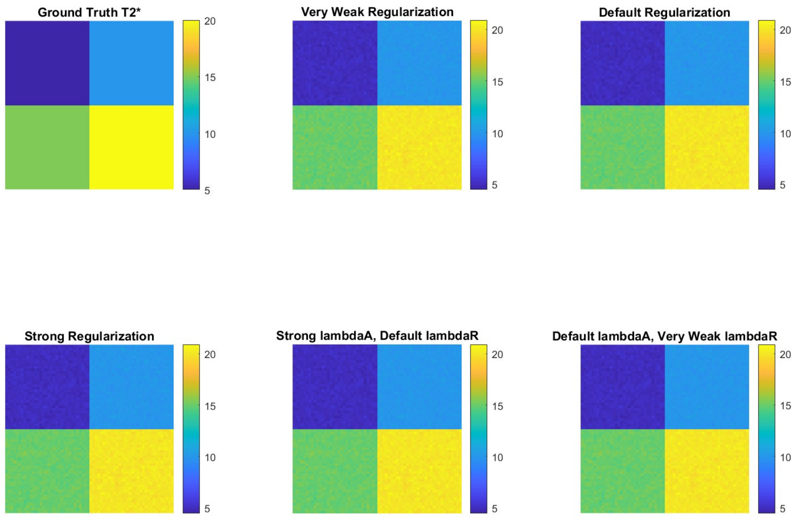


Figure 2: Task 6 - Effect of choice of hyperparameters for regularization on model.

Task 7 - Analysis of MRI Volumes

We loaded and examined the clinical MRI data for iron overload assessment. The patient datasets consisted of 3D image volumes with dimensions 192 $\times$ 256 $\times$ 8, where the third dimension represents the 8 different echo times. The images were acquired at echo times ranging from 2.58 ms to 18.88 ms, with approximately equal intervals between consecutive echoes.

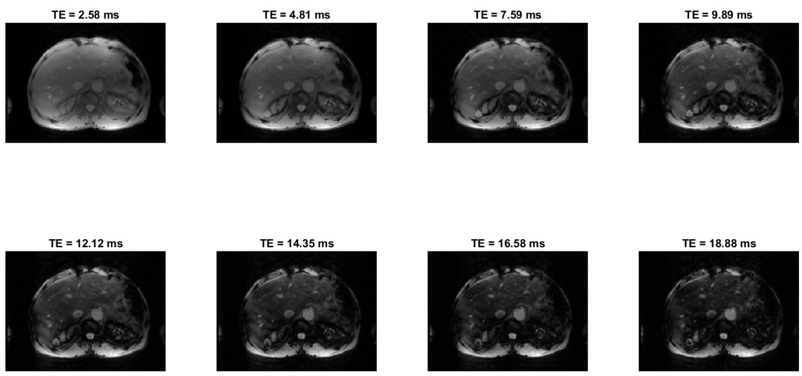


Figure 3: Task 7 - Sequence of DICOM MRI images from patient 2_1 with TE values for each.

Task 8 - Signal Intensity vs. Echo Time Analysis

To analyze the signal decay characteristics, we selected regions of interest (ROIs) in the liver parenchyma of two patients with different iron loading levels. We plotted the mean signal intensity within these ROIs against the echo times, as shown in Figure 4. The decay curves reveal distinct patterns between the two patients. Patient 1_2 (blue curve) shows a higher initial signal intensity (S0 = 250) and a

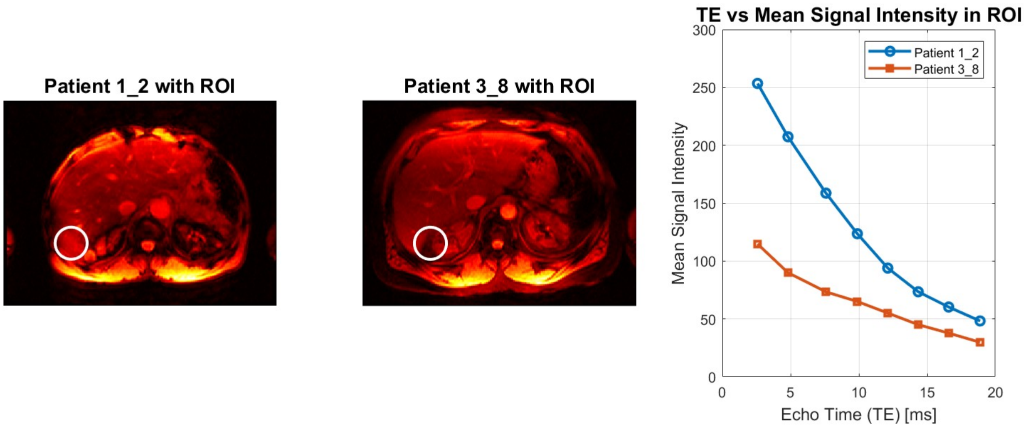


Figure 4: Task 8 - TE vs Mean Signal Intensity in bottom left ROI for patients 1_2 and 3_8, note that patient 3_8's S0 is much lower.

more gradual signal decay, indicative of lower iron concentration. In contrast, Patient 3_8 (red curve) exhibits both a lower initial signal intensity (S0 = 115) and a steeper decay rate, suggesting higher hepatic iron content. This comparison illustrates how the T2* decay curve characteristics directly reflect differences in iron loading between patients, with faster decay corresponding to increased iron deposition.

Task 9 - Clinical Validation

We applied our ADMM-based T2* estimation algorithm to MRI data from four patients and compared our results with those provided by the MRI scanner console. Figure 5 shows the estimated initial signal intensity maps (left) and the corresponding T2* maps (right) for each patient, with numerical results reported above each set of images. Our model produced T2* estimates that generally follow the same pattern as the console values, with Patient 2 showing the lowest T2* values (7.08 ms estimated vs. 4.26 ms console).

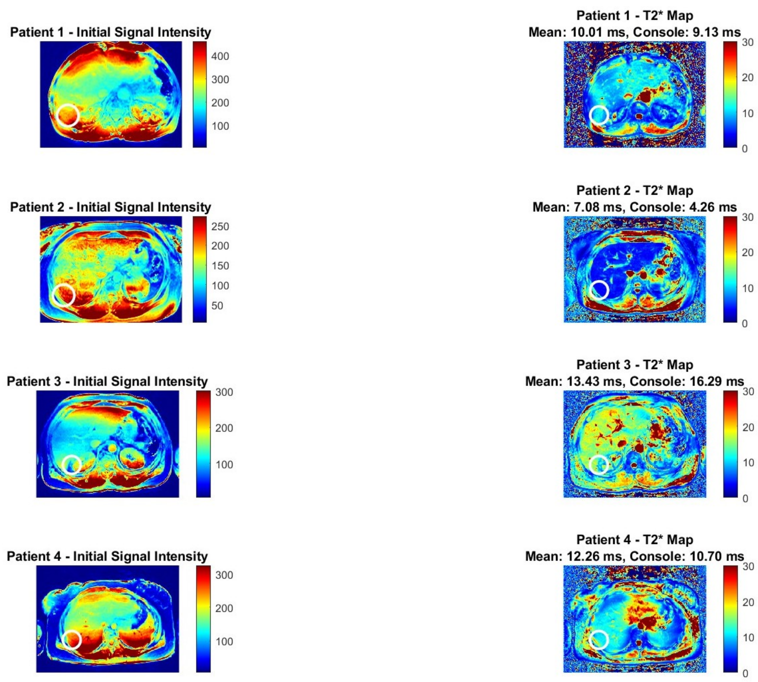


Figure 5: Task 9 - Left images show initial signal intensity for each patient, and right the T2* estimate. Mean estimate for the ROI and Console T2* are also reported above left images.

Task 10 - Comparison with Levenberg-Marquardt Algorithm

LM combines the Gauss-Newton method with gradient descent to solve non-linear least squares problems. For T2* estimation, LM iteratively adjusts the model parameters (initial intensity a and relaxation rate r) to minimize the difference between observed signal intensities and those predicted by the exponential model.

The key difference between LM and our ADMM-based approach lies in how spatial information is utilized. LM processes each voxel independently, treating the T2* estimation as separate curve-fitting problems without considering relationships between neighboring voxels. In contrast, our ADMM method incorporates total variation regularization to enforce spatial coherence, allowing information from adjacent voxels to inform the estimation process.

Task 11 - Alternative Model Implementation

To provide a comprehensive comparison, we implemented the widely used Levenberg-Marquardt (LM) algorithm for T2* estimation and evaluated it against our ADMM approach using both synthetic and clinical datasets.

Table 3: Comparison of ground truth and LM-estimated parameters for synthetic data

Quadrant	True a	Est. a	True T2*	Est. T2*
1	155.00	155.04	5.00	5.01 (0.17%)
2	255.00	254.94	10.00	10.01 (0.09%)
3	355.00	355.04	15.00	15.00 (0.01%)
4	455.00	455.10	20.00	19.99 (0.03%)
			Mean percentage error:	**0.07%**

The LM algorithm demonstrated excellent performance on synthetic data with minimal error. As shown in Table 3 and Figure 6(a), the algorithm accurately recovered both the initial intensity values and T2* parameters across all quadrants, with a mean T2* estimation error of only 0.07%.

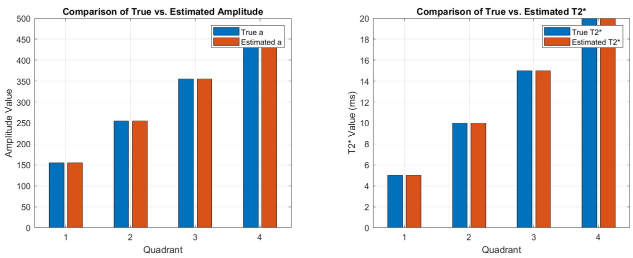
Table 4: Comparison of T2* estimation methods on patient data

Patient	LM T2* (ms)	ADMM T2* (ms)	Console T2* (ms)
1	9.97	10.01	9.13
2	7.04	7.08	4.26
3	13.25	13.43	16.29
4	11.99	12.26	10.70
Mean LM vs. Console Absolute Error: 1.99 ms			
Mean ADMM vs. Console Absolute Error: 2.03 ms			
Mean LM vs. ADMM Absolute Error: 0.13 ms			

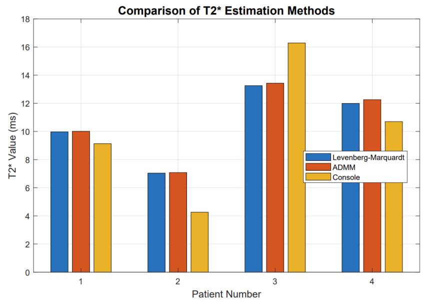
When applied to clinical data, the LM algorithm yielded T2* estimates that were remarkably similar to our ADMM approach (Table 4 and Figure 6(b)). The mean absolute difference between the two methods was only 0.13 ms, indicating that both techniques produce consistent results. Both methods showed similar deviations from the console values.

5 Conclusion

Our experimental results on both synthetic and clinical datasets demonstrate that the proposed method achieves high accuracy in estimating T2* values, with errors below 0.25% on synthetic data even in the presence of noise. When compared with the widely used Levenberg-Marquardt algorithm, our ADMM-based approach produced similar T2* estimates (mean difference of only 0.13 ms), indicating consistent performance between the methods. The ADMM method showed slightly improved

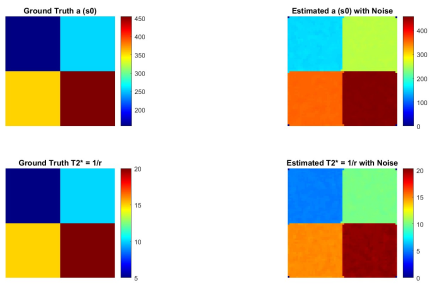


(a)

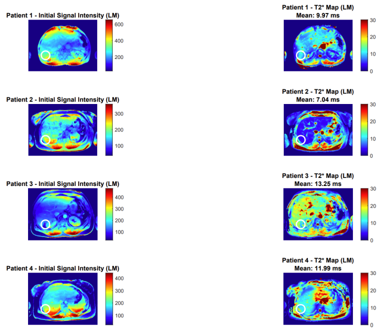


(b)

Figure 6: Task 11: (a) Shows the performance of the LM model on synthetic maps. (b) Compares the Console values, LM, and our proposed ADMM model T2* estimation. LM and ADMM generally perform the same and there is no indication of ADMM being more accurate. Notably, LM and ADMM both fail to accurately model the noise in the high LIC patient 2 (3_8).



(a)



(b)

Figure 7: Task 11: (a) Ground truth and estimated LM synthetic maps with noise. (Note there are a few artifacts in the estimated images, black pixels). (b) Initial and T2* Maps from LM model fitting. Patient 2 again shows strong LIC (deep blue). These maps are generally comparable to ADMM, but LM shows increased noise around circulatory liver regions.

noise handling in vascular regions due to its spatial regularization. However, the computational efficiency of the two methods differed substantially - the ADMM approach required approximately 13 minutes per patient on a Ryzen 7 8-core CPU, while the LM method completed in just a few minutes. This computational cost represents a significant practical limitation for the ADMM method in clinical settings. Both methods showed similar deviations from console-provided T2* values, particularly in cases of high iron concentration. Overall, while our TV-regularized ADMM approach offers theoretical advantages for spatial coherence, the Levenberg-Marquardt algorithm provides a more practical solution for clinical T2* mapping with comparable accuracy and substantially faster processing times.

6 Future Work

Alright, let's do **part (b)** carefully, expanding the sum **directly** as you said, using the suggested trick:

We want to prove that

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

is a **biased** estimator of σ^2.

—

First, recall:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Now, the trick you're asking to use is to rewrite:

$$X_i - \hat{\mu} = (X_i - \mu) - (\hat{\mu} - \mu)$$

Thus:

$$(X_i - \hat{\mu})^2 = ((X_i - \mu) - (\hat{\mu} - \mu))^2$$

Expand the square:

$$= (X_i - \mu)^2 - 2(X_i - \mu)(\hat{\mu} - \mu) + (\hat{\mu} - \mu)^2$$

Now, sum over $i = 1$ to n:

$$\sum_{i=1}^{n}(X_i - \hat{\mu})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - 2(\hat{\mu} - \mu)\sum_{i=1}^{n}(X_i - \mu) + n(\hat{\mu} - \mu)^2$$

Thus:

$$\hat{\sigma}^2 = \frac{1}{n}\left(\sum_{i=1}^{n}(X_i - \mu)^2 - 2(\hat{\mu} - \mu)\sum_{i=1}^{n}(X_i - \mu) + n(\hat{\mu} - \mu)^2\right)$$

Now, take expectations:
Step 1: Expectation of $\sum_{i=1}^{n}(X_i - \mu)^2$
Since the X_i are i.i.d. with variance σ^2,

$$E\left[(X_i - \mu)^2\right] = \sigma^2$$

thus

$$E\left[\sum_{i=1}^{n}(X_i - \mu)^2\right] = n\sigma^2$$

—

Step 2: Expectation of $(\hat{\mu} - \mu)$
Since $\hat{\mu}$ is an unbiased estimator of μ,

$$E[\hat{\mu} - \mu] = 0$$

Thus, $E\left((\hat{\mu} - \mu)\sum_{i=1}^{n}(X_i - \mu)\right)$ simplifies nicely:
First, notice $\sum_{i=1}^{n}(X_i - \mu)$ is just $n(\hat{\mu} - \mu)$, so:

$$(\hat{\mu} - \mu)\sum_{i=1}^{n}(X_i - \mu) = n(\hat{\mu} - \mu)^2$$

Thus, the second term becomes:

$$-2(\hat{\mu} - \mu)\sum_{i=1}^{n}(X_i - \mu) = -2n(\hat{\mu} - \mu)^2$$

—

Step 3: Expectation of $(\hat{\mu} - \mu)^2$
We know that:

$$E[(\hat{\mu} - \mu)^2] = \mathrm{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$$

—

Putting it all together:
Substituting into the formula:

$$
\begin{aligned}
E[\hat{\sigma}^2] &= \frac{1}{n}\left(n\sigma^2 - 2nE[(\hat{\mu}-\mu)^2] + nE[(\hat{\mu}-\mu)^2]\right) \\
&= \frac{1}{n}\left(n\sigma^2 - nE[(\hat{\mu}-\mu)^2]\right) \\
&= \sigma^2 - E[(\hat{\mu}-\mu)^2] \\
&= \sigma^2 - \frac{\sigma^2}{n} \\
&= \sigma^2\left(1-\frac{1}{n}\right)
\end{aligned}
$$

Thus,

$$E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$$

which shows that $\hat{\sigma}^2$ is **biased downward**: it underestimates the true variance by a factor of $\frac{n-1}{n}$.

—

Conclusion:

$$E[\hat{\sigma}^2] \neq \sigma^2$$

specifically, $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$.

—

Would you also like me to quickly show how the *unbiased* estimator is derived from this (i.e., replacing n by $n-1$)? It's very short.

Several promising directions could further enhance the utility of our proposed approach. Deep learning could automate ROI selection through liver segmentation models, eliminating inter-observer variability and reducing processing time. Such models could be trained to identify and exclude vascular structures and bile ducts that may otherwise contaminate T2* measurements (Tipirneni-Sajja et al., 2019). Beyond ROI selection, exploring more sophisticated signal decay models within our ADMM framework could improve accuracy in cases of high iron concentration. Implementing the bi-exponential model to account for both iron-dense and iron-sparse components, or the shifted single-exponential model with a constant offset to handle long T2* components, might better characterize heterogeneous iron distribution within liver tissue.

7 References

References

[1] Tipirneni-Sajja, A., Loeffler, R.B., Ley-Zaporozhan, J., McCarville, M.B., Hankins, J.S., and Hillenbrand, C.M. (2019). Ultrashort echo time imaging for quantification of hepatic iron overload: Comparison of acquisition and fitting methods via simulations, phantoms, and in vivo data. Magnetic Resonance in Medicine, 82(3):1094-1106.

[2] Reeder, S.B., Yokoo, T., França, M., Hernando, D., Alberich-Bayarri, Á., Alústiza, J.M., Gandon, Y., Henninger, B., Hillenbrand, C., Jhaveri, K., Karçaaltıncaba, M., Kühn, J.P., Mojtahed, A., et al. (2023). Quantification of Liver Iron Overload with MRI: Review and Guidelines from the ESGAR and SAR. Radiology, 307(2):e221856.

[3] Hernando, D., Levin, Y.S., Sirlin, C.B., and Reeder, S.B. (2014). Quantification of liver iron with MRI: state of the art and remaining challenges. Journal of Magnetic Resonance Imaging, 40(5):1003-1021.

[4] Ibrahim, E.S., Khalifa, A., Eldaly, A.K. (2016). Influence of the analysis technique on estimating hepatic iron content using MRI. Journal of Magnetic Resonance Imaging, 44(6):1448-1455.

[5] Siegelman, E.S., Mitchell, D.G., Rubin, R., Hann, H.W., Kaplan, K.R., Steiner, R.M., Rao, V.M., Schuster, S.J., Burk, D.L., Jr, and Rifkin, M.D. (1991). Parenchymal versus reticuloendothelial iron overload in the liver: distinction with MR imaging. Radiology, 179(2):361-366.

[6] Eldaly, A.K., and Khalifa, A.M. (2023). A general framework for hepatic iron overload quantification using MRI. Digital Signal Processing, 137:104048.